Jihwan Kim

tho3ov@gmail.com (Personal & Consultant) | jkim@groupcls.com (Company)

EDUCATION

Ph.D. in Physical Oceanography **2019–**

School of Earth and Environmental Sciences, College of Natural Sciences, Seoul National Univ. **2023**

Graduated as the youngest and fastest to complete a Ph.D. in Physical Oceanography in Korea

B.S. in Earth Science Education **2013–**

College of Education, Seoul National University **2018**

**Compulsory Military Service in 2015–2016

RESEARCH INTERESTS

- Pelagic fisheries and upper-ocean current systems in the Northwestern Pacific
- Statistical forecasting methods for fisheries and environmental data
- Machine learning applications for sustainable fishing practices, including super-resolution of coarse spatial-temporal data and forecasting future fishing grounds

SKILLS

Languages:

Native Korean; Professional Working Proficiency in English and Japanese.

Technical Proficiencies:

- **Core Programming & Scientific Computing:**
  - **Advanced Python:** for complex data manipulation, modeling, and analysis.
  - **Advanced MATLAB:** for oceanographic data processing and algorithm development.
  - **Proficient in Julia & Fortran:** Experience in leveraging for high-performance scientific computations.
- **Data Analysis, Modeling & Visualization:**
  - **Spatio-temporal Data Analysis:** Specialized in techniques like EOF and CSEOF for oceanographic and fisheries data.
  - **Statistical Forecasting & Regression Analysis:** Applying predictive models to environmental and fisheries datasets.
  - **Machine Learning:** Development of time-series prediction models using TensorFlow and PyTorch for applications like fishing ground forecasting and data super-resolution.
  - **Geospatial Information Systems (GIS):** Proficient in GIS software for mapping and spatial analysis of catch/effort data.

- **Data Visualization:** Creating informative visualizations from complex datasets

- **Data Management & Web Technologies:**
  - **Database Management:** Proficient with relational databases (MySQL) for data storage, querying, and warehousing.
  - **Dashboard Development:** Experience in creating interactive data dashboards
  - **Front-End Familiarity:** Exposure to front-end frameworks (e.g., Vue, React) for specific project components or data presentation layers.
- **General Applications:**
  - Proficient with MS Office Suite (Word, Excel, PowerPoint) for reporting and presentations.

TEACHING EXPERIENCE

- **Teaching Assistant at Seoul National University**

  Physical Oceanography (Spring 2019)

  Observation and Comprehension of Natural Disasters (Fall 2019, Spring 2020)

  Theory and Practice in Computational Sciences 1 (Spring 2022)

- **Lectures at Seoul National University**

  Computational Science: Subject Research (The Missing Semester of CS Education) (Spring 2021)

HONORS AND AWARDS

- **JOISS Marine Science Big Data Competition** **2021–2022**

  2nd Prize rewarded by Chairman of the Korea Oceanographic Society (2021 and 2022, respectively). JOISS Marine Science Big Data Competition is selected as the UN Ocean Decade Activity.

- **Succeeding Generations of Fundamental Sciences Fellowship** **2019–2020**

  Selected in the field of Oceanography. Seoul National University awarded a full scholarship to graduate students who can be a succeeding leader of education and research in fundamental sciences.

- **Travel Grants for Young Scientists**

  PICES 30th annual meeting in Busan, Republic of Korea (2022)

EXPERIENCE

- **Ocean Engineer, Sustainable Fisheries Management Team, Collecte Localisation Satellites (CLS) – France & Japan** **2025–**
  - **RFMO LL Favorability Map:** Developed five longline models for yellowfin tuna, bigeye tuna, and swordfish in the Indian and Pacific Oceans. Combined monthly RFMO catch and effort records with daily surface and subsurface ocean data to generate daily favorability rankings at 1/12° resolution. Evaluated the maps against historical catch per unit effort at the original monthly 5° RFMO scale.
  - **ENSO/IOD Catch Alert:** Developed statistical and machine-learning models that use climate signals to anticipate changes in tuna purse-seine catches up to six months ahead.
  - Develop oceanographic data integration algorithms and define data requirements for predictive features in CATSAT, a satellite-based fisheries platform.
- **Data & Fisheries Science Advisor to the North Pacific Fisheries Commission** **2024–**
  - **Resource Management - Pacific Saury**:
    - Delivered presentations and prepared publications on the spatial-temporal variability in catch per unit effort of Pacific saury (*Cololabis saira*) and its relationship with interannual basin-scale environmental variability in the North Pacific.
    - Presented research on stochastic statistical simulations for forecasting the spatial-temporal distribution of Pacific saury based on observational data in the Northwestern Pacific.
    - Developed and presented the "Physically Consistent Super-Resolution of Pacific Saury Spatio-Temporal Distribution Using Multi-head Attention U-Net."
  - **Resource Management - Chub Mackerel**:
    - Investigated and reported on the effects of Kuroshio Current variability and the Pacific Decadal Oscillation on the recent decline in Chub Mackerel (*Scomber japonicus*) catch in the Northwestern Pacific during the 2020s.
  - **GIS & Data Integration**:
    - Developed a GIS-based mapping system to visualize and analyze Neon Flying Squid catch and effort data submitted by North Pacific Fisheries Commission members.
  - **Compliance Analytics & Reporting**:

- Designed matching algorithms to integrate Vessel Monitoring System (VMS) data with submitted transshipment documents, high seas boarding inspection reports, and Automatic Identification System data, thereby detecting potential non-compliance.
- Produced and presented comprehensive reports on Fisheries Overview, Transshipment Overview, and VMS Overview at North Pacific Fisheries Commission Meetings in Osaka (2024 and 2025).
- Compiled the 2024 North Pacific Fisheries Commission Fisheries and Marine Resources Inventories for inclusion in the Food and Agriculture Organization's Fisheries and Resources Monitoring System.

- **Postdoctoral Fellowship in the North Pacific Fisheries Commission and Tokyo University of Marine Science and Technology** **2023–2024**
  - **Resource Management - Pacific Saury**:
  - Prepared a presentation and publication titled "The Interannual to Decadal Relationship Between Total Catch Variability of Pacific Saury (*Cololabis saira*) and Basin-Scale Ocean Environmental Variability in the North Pacific."
  - **Resource Management - Chub Mackerel**:
  - Prepared a presentation and publication on "Assessment of the Interannual to Decadal Relationship Between Catch and Population Variability of Chub Mackerel (*Scomber japonicus*) in Japan and Basin-Scale Ocean Environmental Variability in the North Pacific."
  - **Catch Reporting Measures**:
  - Developed strategic measures for reprinting catches to improve accuracy in fisheries reporting.
  - **GIS-Based Mapping**:
  - Designed and implemented a GIS-based mapping system for Neon Flying Squid catch and effort data from North Pacific Fisheries Commission members.
  - **IUU Fishing Measures**:
  - Developed algorithms for the Vessel Monitoring System to detect IUU fishing vessels.
  - Created algorithms to integrate and match air surveillance data with VMS data for enhanced detection and monitoring.

- **High school Science Teacher** **2018–2019**

  Earth Science Teacher at Sehwa High School, Seoul, Republic of Korea

- **Auxiliary Policeman (Compulsory Military Service)** **2015–2016**

  Strike Force of the Police in Gangwon Province, Republic of Korea

EXPERIENCE IN FUNDED PROJECT

- **A study on upstream-downstream connectivity of the western boundary current in the North Pacific and its relationship with mesoscale eddy activity** **2019–2023**

  funded by the National Research Foundation of the Republic of Korea

  investigates the Kuroshio current variability and its ecological impact variability

- **Seychelles-Chagos Thermocline Ridge Pilot Time-series Observation Initiative**

  funded by the Korea Institute of Ocean Science Technology

  investigates upwelling and related ecological variability in Seychelles-Chagos Thermocline Ridge

- **East Asian Seas Time-series** **2021– 2023**

  funded by the Ministry of Land, Transport and Maritime Affairs, Korea

  investigates ocean-environmental variability (physical, biological, etc.) in the East/Japan Sea

FIELD EXPERIENCE

- **The East China Sea and Yellow Sea (CTD, ADCP, DO sampling)**

  R/V ONNURI (Jul 2022)

- **The Korea/Tsushima Strait (Induced voltage measurement on submarine cable)**

  Submarine cable voltage measurement between Busan, Republic of Korea and Hamada, Japan. Volume transport through the Korea/Tsushima Strait can be estimated from the motion-induced voltage on submarine cable across the strait. Submarine cable voltage observations were stopped on May 17, 2017 owing to a measurement equipment failure and re-initiated on Dec 13, 2020. Jihwan Kim devised and built a new observation solution, a new data analysis method, and a cloud data server.

PRESENTATIONS

- **2024 Vessel Monitoring System Overview** **Mar, 2025**

  8th TCC Meeting, Osaka, Japan, Mar 20, 2025

- **2024 Transshipment Overview**

  8th TCC Meeting, Osaka, Japan, Mar 19, 2025

- **Revised GIS-based mapping system for Neon Flying Squid (*Ommastrephes bartramii*) catch and effort data from members of the North Pacific Fisheries Commission** **Dec, 2024**

  14th SSC PS meeting, Tokyo, Japan, Dec 18, 2024

- **Spatial-temporal variability in catch per unit effort of Pacific saury (*Cololabis saira*) and its relationship to interannual basin-scale environmental variability in the North Pacific** **Aug, 2024**
  13th SSC PS meeting, Online, Aug 27, 2024
- **Effects of Kuroshio Current Variability and Pacific Decadal Oscillation on Recent Decline in Chub Mackerel (*Scomber japonicus*) Catch in the Northwestern Pacific in the 2020s** **Jul, 2024**
  9th TWG CSMA Meeting, Yokohama, Japan, Jul 19, 2024
- **Vessel Monitoring System Overview from 2022 to 2023** **Apr, 2024**
  7th TCC Meeting, Osaka, Japan, Apr 9, 2024
- **A GIS-based mapping system for Neon Flying Squid (*Ommastrephes bartramii*) catch and effort data from members of the North Pacific Fisheries Commission** **Dec, 2023**
  12th SSC PS meeting, Nanaimo, Canada, Dec 18, 2023

- **The interannual to decadal relationship between total catch variability of Pacific saury (*Cololabis saira*) and basin-scale ocean environmental variability in the North Pacific** **Dec, 2023**
  12th SSC PS meeting, Nanaimo, Canada, Dec 12, 2023

- **Monitoring volume transport through measurement of cable voltage across the Korea Strait Since November 2020** **Nov, 2022**
  Korea Oceanographic Society Autumn Conference, Gangneung, Republic of Korea, Nov 2, 2022
- **Interannual variability of barotropic sea level difference across the Korea/Tsushima Strait and its relationship to upper-ocean currents variability in the western North Pacific** **Sep, 2022**
  PICES-2022 annual meeting, Busan, Republic of Korea, Sep 30, 2022

- **Interannual variability of barotropic sea level difference across the Korea/Tsushima Strait and its relationship to upstream Kuroshio variability** **Mar, 2022**
  Ocean Sciences Meeting, Online Everywhere, Mar 3, 2022
- **Interannual variability of Yellowfin tuna (*Thunnus albacares*) and Bigeye tuna (*Thunnus obesus*) catches in the southwestern tropical Indian Ocean and its relationship to climate variability** **Dec, 2020**
  AGU Fall Meeting, Online Everywhere, Dec 7, 2020

- **Potential Predictability of Skipjack Tuna (*Katsuwonus pelamis*) Catch in the Western Central Pacific** **Oct, 2019**
  Korea Oceanographic Society Autumn Conference, Gangneung, Republic of Korea, Oct 31, 2019

PUBLICATIONS

- **Jihwan Kim** and Hanna Na (2022), Interannual Variability of Yellowfin Tuna (*Thunnus albacares*) and Bigeye Tuna (*Thunnus obesus*) Catches in the Southwestern Tropical Indian Ocean and Its Relationship to Climate Variability, *Frontiers in Marine Science*, **9**, 857405
- **Jihwan Kim**, Hanna Na, Young-Gyu Park, and Young Ho Kim (2020), Potential Predictability of Skipjack Tuna (*Katsuwonus pelamis*) Catches in the Western Central Pacific, *Scientific Reports*, **10**, 3193.

MANUSCRIPT SUBMITTED

- **Jihwan Kim**, Libin Dai, Christopher N. Rooper, Yi-Jay Chang, Jhen Hsu, Hyejin Song, Vladimir Kulik, Mei-chin Juan, Robert Day, Aleksandr Zavolokin, Rentaro Mitsuyu, and Toshihide Kitakado, Spatio-temporal variability of density distribution of Pacific saury (Cololabis saira) and its relationship to basin-scale Ocean environmental variability in the North Pacific
- **Jihwan Kim**, Christopher N. Rooper, Shota Nishijima, Kazuhiro Oshima, Robert Day, and Aleksandr Zavolokin, Effects of Kuroshio Current Variability and Pacific Decadal Oscillation on Recent Decline in Chub Mackerel (*Scomber japonicus*) Catch in the Northwestern Pacific in the 2020s

MANUSCRIPT IN PREPARATION

- **Jihwan Kim**, Libin Dai, Christopher N. Rooper, Shinichiro Nakayama, Yi-Jay Chang, Jhen Hsu, Hyejin Song, Vladimir Kulik, Me-chin Juan, Robert Day, Aleksandr Zavolokin, Rentaro Mitsuyu, and Toshihide Kitakado, Stochastic statistical simulations of spatial-temporal distribution of Pacific saury (*Cololabis saira*) and its forecast based on observational data in the Northwestern Pacific
- **Jihwan Kim** and Hanna Na, Assessment of the Favorable Ocean Conditions and Potential Predictability of Annual Catch amounts of Skipjack tuna (*Katsuwonus pelamis*), Yellowfin tuna (*Thunnus albacares*), and Bigeye Tuna (*Thunnus obesus*) in the Western Central Pacific